

1) a) Overall DOF: $14 + 2 - (1 + 1 + 1 + 14) = -1$

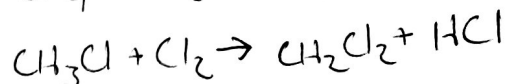
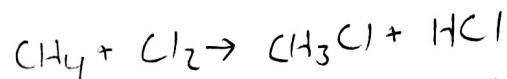
Mixer DOF: $3 + 0 - (1 + 1 + 0 + 2) = -1$

Reactor DOF: $6 + 2 - (1 + 1 + 1 + 4) = -3$

Condenser DOF: $8 + 0 - (0 + 0 + 1 + 4) = 3$

Absorber DOF: $4 + 0 - (0 + 0 + 0 + 2) = 2$

Still DOF: $4 + 0 - (0 + 0 + 0 + 2) = 2$



b) Reactor Balance

CH_4 : $n_3 = 80 - \dot{g}_1$

~~CH_2Cl_2 : $n_3 = \dot{g}_2$~~

Cl_2 : $0 = 20 - \dot{g}_1 - \dot{g}_2 \Rightarrow 20 - \dot{g}_1 = \dot{g}_2 \Rightarrow 20 - 10.9 = 9.1$

CH_3Cl : $5 \cdot n_5 = \dot{g}_1 - \dot{g}_2 \Rightarrow n_5 = \frac{\dot{g}_1 - \dot{g}_2}{5} = \frac{10.9 - 9.1}{5} = 0.36$

HCl : $n_4 = \dot{g}_1 + \dot{g}_2 \Rightarrow n_4 = 10.9 + 9.1 = 20$

Total: $n_3 + n_4 + 6n_5 = 100$

$80 - \dot{g}_1 + 20 + 6 \left(\frac{2\dot{g}_1 - 20}{5} \right) = 100 \Rightarrow \frac{11\dot{g}_1 - 120}{5} = 0 \Rightarrow 11\dot{g}_1 = 120 \Rightarrow \dot{g}_1 = 10.9$

Mixer balance:

CH_4 : $\dot{g}_3 + n_1 = 80 \Rightarrow n_1 = 10.9$

Cl_2 : $n_2 = 20$

$\dot{g}_1 = 10.9$	$n_1 = 10.9$
$\dot{g}_2 = 9.1$	
$n_3 = 69.1$	69.1
$n_4 = 20$	
$n_5 = 1.82$	
$5n_5 = 9.1$	9.1

↓
14.

c) $1000 \frac{\text{kg}}{\text{h}} \times \frac{1 \text{ kmol}}{50 \text{ kg}} \Rightarrow 20 \frac{\text{kmol}}{\text{h}}$ of $\text{CH}_3\text{Cl} = 5n_5$

$n_5 = 4$

$\frac{4}{1.82} = 2.20$ conversion

$\frac{24}{40}$

new $n_3 = 69.1 \cdot 2.2 = 152.02 \frac{\text{kmol}}{\text{h}}$

new $n_2 = 20 \cdot 2.2 = 44 \frac{\text{kmol}}{\text{h}}$

new $n_1 = 10.9 \cdot 2.2 = 23.98 \frac{\text{kmol}}{\text{h}}$

Scale Factor = 1.37

-5

1) 24
2) 30
3) 36

$\frac{84}{100}$

$\dot{n}_1 = 100 \frac{\text{gmol}}{\text{min}}$
 $70\% \text{ M}$
 $30\% \text{ E}$

Mixer $\dot{n}_2$

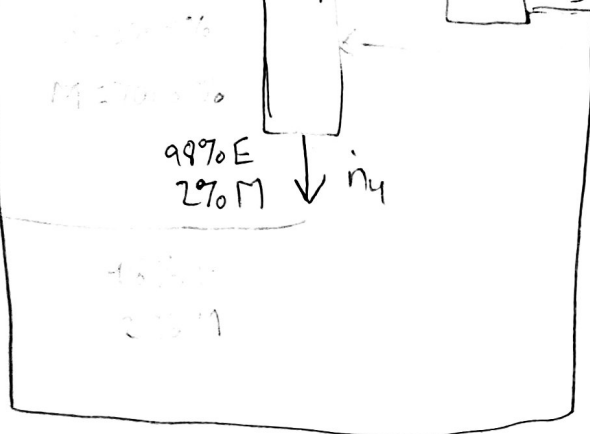
$90\text{mol}\% \text{ M}$
 $10\text{mol}\% \text{ E}$

$\dot{n}_3$

$97\text{mol}\% \text{ M}$
 $3\text{mol}\% \text{ E}$

$\dot{n}_5 = 100 \frac{\text{gmol}}{\text{min}}$

methanol = M
 ethanol = E



$E = ?$
 $M = ?$

	$x_{s,M}$	$x_{z,M}$	$x_{z,E}$	$x_{s,E}$	$\dot{n}_2$	$\dot{n}_3$	$\dot{n}_4$	$\dot{n}_6$
1	✓	✓			✓			
2			✓	✓	✓			
3		✓			✓	✓	✓	
4			✓		✓	✓	✓	
5	✓						✓	✓
6				✓		✓		✓
7	✓			✓				
8		✓	✓					

mixer:

M: $70 + x_{s,M}(100) = x_{z,M} \cdot \dot{n}_2$

E: $30 + x_{s,E}(100) = x_{z,E} \cdot \dot{n}_2$

Dist 1:

M: $x_{z,M} \cdot \dot{n}_2 = 0.90 \cdot \dot{n}_3 + 0.02 \cdot \dot{n}_4$

E: $x_{z,E} \cdot \dot{n}_2 = 0.10 \cdot \dot{n}_3 + 0.98 \cdot \dot{n}_4$

$70 + x_{s,M}(100) = 0.90 \cdot \dot{n}_3 + 0.02 \dot{n}_4$ ⑨

$30 + x_{s,E}(100) = 0.10 \cdot \dot{n}_3 + 0.98 \dot{n}_4$ ⑩

using ③⑥⑦⑨⑩ 5 eqns, 5 unknowns

solved in excel

Dist 2:

M: $0.90 \cdot \dot{n}_3 = 0.97 \cdot \dot{n}_6 + x_{s,M} \cdot 100$

E: $0.10 \dot{n}_3 = 0.03 \cdot \dot{n}_6 + x_{s,E} \cdot 100$

⑦ $x_{s,M} + x_{s,E} = 1$

⑧ $x_{z,M} + x_{z,E} = 1$

$x_{s,M} = 0.06$
 $x_{s,E} = 0.94$

recycle comp

$\dot{n}_3 = 84.1$

$\dot{n}_4 = 28.4$

$\dot{n}_6 = 71.6$ final product

$f_{R,M,6} = \frac{0.97 \cdot 71.6}{70} = 0.99$

$f_{R,E,4} = \frac{0.98 \cdot 28.4}{30} = 0.93$

$\frac{30}{30}$

$$n_1 = 100 \frac{\text{kmol}}{\text{h}}$$

basis

30% E

70% W

$$n_2 = 50 \frac{\text{kmol}}{\text{h}} \text{ Vapor}$$

$$Y_E = 0.4704$$

$$Y_W = 0.5296$$

Tank
1 atm

Liquid

$$n_3 = 50 \frac{\text{kmol}}{\text{h}}$$

$$X_E = 0.1238$$

$$X_W = 0.8762$$

$$30 = Y_E(50) + X_E(50)$$

$$70 = Y_W(50) + X_W(50)$$

or 80.7°C:

when $Y_E + X_E = 0.60$, equation is satisfied

$$\text{At } 85.3^\circ\text{C} \quad \overset{X_E}{0.1238} + \overset{Y_E}{0.4704} = 0.5942$$

85.3°C is very close

$$X_W = 1 - 0.1238 = 0.8762$$

$$Y_W = 1 - 0.4704 = 0.5296$$

$$F_{RE2} = \frac{0.4704(50)}{30} = 0.784$$

$$F_{RW2} = \frac{0.5296(50)}{70} = 0.378 \rightarrow FRW2 = 6.626$$

$$\alpha_{EW} = \frac{0.4704}{0.5296} \cdot \frac{0.8762}{0.1238} = 6.29$$

$$\frac{30}{30}$$

Mathematical - M
relation Gibbs